WEEK 2 WEB APPLICATION INTERNSHIP PROJECT

Responsive Portfolio Website

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Technologies
HTML5 · CSS3 · JavaScript

Project type
Frontend website demonstration

Submission period
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A beginner-friendly project demonstrating page structure, styling, responsive layout, and simple browser-side interactivity.

1. Introduction

This project is a responsive portfolio-style website created for the Week 2 Web Application Internship task. The purpose is to practise the three core frontend technologies: HTML, CSS, and JavaScript. The website includes a header and navigation, hero section, about section, service cards, skills section, contact form, call-to-action area, and footer.

The design uses a blue-and-white visual theme, clear spacing, card layouts, and small animations. It is intended as a learning demonstration and can be personalised before submission.

2. Project Objectives

The objectives of the project are to build a semantic page structure, apply basic CSS styling and layout techniques, add interactive behaviour with JavaScript, practise responsive design, and organise the files into a maintainable project folder.

Expected outcomes

• A multi-section website built with semantic HTML.
• CSS styles using Flexbox, Grid, hover effects, and media queries.
• JavaScript interactions for a button, mobile navigation, and form validation.
• A clear project structure that is easy to open and test in VS Code.

3. Tools and Technologies

HTML5: Provides the content and semantic structure of the website. Elements such as header, nav, main, section, article, form, and footer help organise information meaningfully.

CSS3: Controls the visual appearance, including typography, colours, spacing, borders, shadows, hover effects, and responsive layouts. Flexbox is used for aligned rows and navigation, while CSS Grid is used for larger page layouts and cards.

JavaScript: Adds browser-side interaction. The script displays an alert when the greeting button is clicked, opens and closes the mobile navigation, checks form input, and updates the copyright year.

VS Code and browser: VS Code is used to edit the project files. A modern browser is used to view the page and test the interactions.

4. HTML Structure

The index.html file contains the complete website structure and links the separate style.css and script.js files. The page uses a viewport meta tag to support mobile devices and a description meta tag to describe the project. Navigation links point to section IDs so users can move around the page.

The contact form includes labels, text inputs, an email input, a message textarea, and a submit button. Accessible labels and status feedback help users understand the form controls and validation result.

5. CSS Design and Responsiveness

The style.css file defines reusable colours and common layout rules. A consistent palette uses dark navy text, blue accents, light backgrounds, and subtle borders. Typography, padding, border radius, and shadows are used to create a clean interface.

Flexbox supports the navigation bar, button groups, footer, and call-to-action row. CSS Grid organises the hero, about area, service cards, skills layout, and contact area. Hover effects make cards and buttons respond visually when a pointer is placed over them.

Responsive behaviour is handled by media queries. At narrower widths, the hero and content grids stack vertically, the service cards become a single column, and the navigation is replaced by a toggle button. A reduced-motion rule also limits animations for users who prefer less motion.

6. JavaScript Functionality

Greeting button: Clicking “Say hello” displays a browser alert. This demonstrates an event listener and a simple interaction.

Mobile menu: The menu toggle adds or removes a CSS class and updates its accessible expanded state. Selecting a navigation link closes the menu.

Form validation: The script checks that the name contains at least two characters, the email follows a basic email pattern, and the message contains at least ten characters. Invalid fields receive a visual indicator and a feedback message.

Important limitation: The contact form is a frontend demonstration only. It does not send email or save data to a database because no backend service is connected.

7. Project Folder Structure

The project is organised as follows:

```
Week2-Web-Internship/
■■■ index.html
■■■ style.css
■■■ script.js
■■■ images/
■ ■■■ hero-illustration.svg
■ ■■■ profile-avatar.svg
■ ■■■ README.txt
■■■ screenshots/
■■■ report.pdf
```

The HTML, CSS, and JavaScript are kept in separate files to make the project easier to understand and maintain. The images folder contains SVG illustrations used by the page. The screenshots folder is provided for adding genuine screenshots from the browser after testing.

8. Testing and Results

Recommended manual tests: open index.html in a modern browser; click each navigation link; click “Say hello” and confirm the alert appears; submit the form with empty fields; enter an invalid email; enter a short message; then submit valid sample information and confirm the success message appears. Resize the browser to a mobile width and check that the menu toggle appears and works.

The project is expected to run locally without a server because it uses static HTML, CSS, JavaScript, and SVG assets. An internet connection may be used to load the optional Google Fonts; system fonts remain as fallbacks.

9. Learning Outcomes

By completing and testing this project, a learner can practise semantic HTML, linking stylesheets and scripts, using class and ID selectors, creating layouts with Flexbox and Grid, applying responsive media queries, and handling user events with JavaScript.

The form exercise also introduces basic input validation and user feedback. The project structure reinforces the value of keeping content, design, behaviour, images, and documentation organised.

10. Conclusion

The portfolio website demonstrates the main requirements of the Week 2 frontend task. It includes the requested page sections, visual styling, images, links, lists, hover effects, responsive layout, and multiple JavaScript features. It can be extended with real project links, a personal photograph, social profiles, or a backend contact service in the future.

Before submission: Replace any placeholder contact details, verify the name and project details, open the website in a browser, and capture 2–3 real screenshots of the completed page. Review your internship provider's submission instructions before uploading.